BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Complex Variables & Laplace Transform (MAT215)

Laplace Transform Using Table

First Translation Theorem, Transform of $t^n f(t)$

April 15, 2025

Conducted by

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

For updates subscribe on @ParthoSutraDhor

What is Laplace Table?

Laplace Transform

The Laplace transform of a function $f(t)$, denoted by $\mathcal{L}\{f(t)\}$, is defined as

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt,$$

where s is a complex number and the integral converges.

Linearity of Laplace Transform

Theorem

Let $f(t)$ and $g(t)$ be functions such that $\mathcal{L}\{f(t)\}$ and $\mathcal{L}\{g(t)\}$ exist. Then, for any constants c, we have

$$\mathcal{L}\{cf(t)\} = c\,\mathcal{L}\{f(t)\},$$

$$\mathcal{L}\{f(t) + g(t)\} = \mathcal{L}\{f(t)\} + \mathcal{L}\{g(t)\}.$$

Laplace Transform Using Standard Table

Problem

Evaluate

$$\mathcal{L}\{3e^{2t}\}$$

using the standard table.

Problem

Evaluate

$$\mathcal{L}\{4t^3 - e^{-t}\}$$

using the standard table.

Problem

Evaluate

$$\mathcal{L}\{(t^2+1)^2\}$$

using the standard table.

Problem

Evaluate

$$\mathcal{L}\{7\sin 2t - 3\cos 2t\}$$

using the standard table.

Problem

Evaluate

$$\mathcal{L}\{7\sinh 2t - 3\cosh 2t\}$$

using the standard table.

Important Trigonometric Identities

Compound Angle Identities

$$\begin{aligned}
\sin(A+B) &= \sin A\cos B + \cos A\sin B,\\
\sin(A-B) &= \sin A\cos B - \cos A\sin B,\\
\cos(A+B) &= \cos A\cos B - \sin A\sin B.\\
\cos(A-B) &= \cos A\cos B + \sin A\sin B.
\end{aligned}$$

Product-to-Sum Identities

$$\begin{aligned}
2\sin A\cos B &= \sin(A+B) + \sin(A-B),\\
2\cos A\cos B &= \cos(A+B) + \cos(A-B),\\
2\sin A\sin B &= \cos(A-B) - \cos(A+B).
\end{aligned}$$

Negative Angle Identities

$$\begin{aligned}
\sin(-A) &= -\sin(A),\\
\cos(-A) &= \cos(A).
\end{aligned}$$

Problem

Evaluate

$$\mathcal{L}\{3 \sin 7t \cos 4t\}$$

using the standard table.

Problem

Evaluate

$$\mathcal{L}\{4\cos 3t \cos 5t\}$$

using the standard table.

Problem

Evaluate

$$\mathcal{L}\{4\cos 3t\cos 7t\}$$

using the standard table.

First Translation Theorem

First Translation Theorem

Let $f(t)$ be a function such that $\mathcal{L}\{f(t)\}$ exists and $F(s) = \mathcal{L}\{f(t)\}$. Then, for any constant a, we have

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a).$$

Problem

Evaluate

$$\mathcal{L}\{t^3e^{5t}\}$$

Problem

Evaluate

$$\mathcal{L}\{5e^{-3t}\sin 4t\}$$

Problem

Evaluate

$$\mathcal{L}\{e^{-t}(3\sinh 2t - 5\cosh 2t)\}$$

Problem

Evaluate

$$\mathcal{L}\{2e^{3t}\sin 5t \sin 3t\}$$

Problem

Evaluate

$$\mathcal{L}\{3e^{3t}\cos 5t\cos 7t\}$$

Problem

Evaluate

$$\mathcal{L}\{4e^{3t}\sin 5t\cos 3t\}$$

Laplace Transform of the form $t^n f(t)$

Theorem

Let $f(t)$ be a function such that $\mathcal{L}\{f(t)\}$ exists and $F(s) = \mathcal{L}\{f(t)\}$. Then, for any positive integer n, we have

$$\mathcal{L}\{t^n f(t)\} = (-1)^n F^{(n)}(s),$$

where $F^{(n)}(s)$ denotes the n-th derivative of $F(s)$ with respect to s.

Problem

Evaluate

$$\mathcal{L}\{t^2 \sin 2t\}$$

Problem

Evaluate

$$\mathcal{L}\{te^{-2t}\sin 3t\}$$

Problem

Evaluate

$$\mathcal{L}\{te^{2t}(2\sinh 3t + 5\cosh 4t)\}$$

Problem

Evaluate

$$\mathcal{L}\{3t \sin 7t \cos 5t\}$$

Problem

Evaluate

$$\mathcal{L}\{4t\cos 3t\cos 5t\}$$

Problem

Evaluate

$$\mathcal{L}\{7te^{-4t}\sin 5t \sin 3t\}$$

Thank You!

We'd love your questions and feedback.

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

partho.dhor@bracu.ac.bd | parthosutradhor@gmail.com

@ParthoSutraDhor

(Lectures, walkthroughs, and course updates)

Scan for the channel

References

[1] Murray R. Spiegel, *Schaum's Outline of Complex Variables*, 2nd Edition, McGraw–Hill, 2009.

[2] Dennis G. Zill, *A First Course in Differential Equations with Modeling Applications*, 11th Edition, Cengage, 2018.